CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The resistance of a conductor is 1

(a) inversely proportional to its length **(b)** directly proportional to its length
(c) proportional to the square of the area **(d)** independent of length

2. The equivalent resistance of 2 ohm and 2 ohm in parallel is 1

(a) 4 ohm **(b)** 1 ohm **(c)** 2 ohm **(d)** 0.5 ohm

3. One volt is the potential difference when 1

(a) 1 watt of power is used for 1 s
(b) 1 joule of work is done to move 1 coulomb of charge
(c) 1 ohm of resistance carries 1 A
(d) 1 coulomb of charge moves in 1 s

4. The resistance of a conductor is inversely proportional to its 1

(a) temperature only **(b)** resistivity
(c) area of cross-section **(d)** length

5. The effective resistance of three resistors of 3 ohm each in series is 1

(a) 9 ohm **(b)** 1 ohm **(c)** 3 ohm **(d)** 6 ohm

6. The relation between charge, current and time is 1

(a) Q = I / t **(b)** Q = t / I **(c)** Q = I t **(d)** Q = I + t

7. The SI unit of resistivity is 1

(a) siemens **(b)** ohm metre **(c)** ohm per metre **(d)** ohm

8. The equivalent resistance of parallel resistors is 1

(a) equal to the average **(b)** less than the smallest resistance
(c) more than the largest resistance **(d)** equal to the sum

9. Which of the following is a good conductor of electricity? 1

(a) Copper **(b)** Glass **(c)** Rubber **(d)** Wood

10. Resistivity of an alloy is generally 1

(a) same as that of insulators
(b) lower than that of its metals
(c) zero
(d) higher than that of its constituent metals

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are alloys used in heating devices? 2

12. State Joule's law of heating. 2

13. State Ohm's law. 2

14. Why is copper used for connecting wires? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A wire of resistance 20 ohm carries a current of 0.5 A. Find the potential difference across it. 3

16. State three differences between series and parallel combinations of resistors. 3

17. Why is tungsten used for bulb filaments? 3

OR

Draw the circuit diagram in words: how would you connect a cell, a resistor, an ammeter and a voltmeter to verify Ohm's law? 3

18. A 6 ohm and a 3 ohm resistor are in parallel and connected in series with 4 ohm. Find the total resistance. 3

OR

State the factors on which the heat produced in a conductor depends and write the formula. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Derive the formula for the equivalent resistance of two resistors in parallel. Find it for 12 ohm and 4 ohm. 5

OR

Three resistors of 2 ohm, 3 ohm and 6 ohm are connected in parallel across a 6 V source. Find the total resistance, total current and the power consumed. 5

20. A resistance of 40 ohm and one of 60 ohm are in parallel. A 12 V battery is connected. Find the total resistance and the current in each resistor. 5

OR

Explain the factors affecting resistance with expression, and compare the resistance of copper and nichrome wires of the same size. 5

END OF QUESTION PAPER — Total: 40 Marks

Mastermind